

04 — Design and interpret carbon and alkalinity experiments

Learning goals: interpret how carbon and alkalinity inputs change atm CO₂, ocean chemistry and carbonate preservation; explain critical-depth motion and OA/OAE asymmetry using matched controls.

Provisional time: 40 minutes. Prediction/specification (5), two forcing tasks (10), supplied runs/checks (10), four short answers (15). Three cases are run: control, OA and OAE. All plotting and numerical machinery are supplied. [Teaching goals](#).

Reading key: **Key term** = concept to notice. Blue **Question** panels identify student work; purple **Instructor answer** panels appear only in the instructor sheet. Code labels distinguish **Choose and explain**, **Understand and run**, and **Supplied implementation**. This practical is ungraded. Keep your explanations here; no separate submission is required. The [coding cheatsheet](#) is optional lookup support; essential syntax is explained locally.

LECTURER REVIEW | INSTRUCTOR EDITION

Executed in ESBMTK314. All instructor solutions, code, execution outputs and optional disclosures are included. No Jupyter installation is needed to read this PDF.

Source: 04_pump_strength_OA_OAE.ipynb

Execution: 2026-09-24. The source notebooks are unchanged.

```
# Supplied implementation: run this support code as provided.
from pathlib import Path
import sys
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

ROOT = next(p for p in (Path.cwd().resolve(), *Path.cwd().resolve().parents)
             if (p / 'teaching_config.py').is_file())
if str(ROOT) not in sys.path:
    sys.path.insert(0, str(ROOT))

from esbmtk import Signal, Source, Species2Species
from model import initialize_model, postprocess_carbonate_horizons, run_model
from presets import load_boudreau_parameters, make_pump_variant
from model_inputs import read_model_tables

DATA = ROOT / 'data' / 'Boudreau_2010'
WORKBOOK = DATA / 'model_definition.xlsx'
P = load_boudreau_parameters(WORKBOOK)
input_tables = read_model_tables(WORKBOOK)
STATE = DATA / 'steady_state'
PULSE_FILE = DATA / 'IS92a-scenario.csv'
DIGITIZED = DATA / 'digitized'
REFERENCE_SCALE = 0.877
PULSE_START = 1800.0
REFERENCE CARBON PMOL = 335.3560189847107
from teaching_audits import audit_complete_model
```

A1. Reuse the model specification from 03

Use your completed 03 schematic as the experiment map; add the external input arrow. Each case uses an independent copy of the same workbook geometry, box-specific T/S/P, transport and baseline rates, and the same saved nearly stationary state (the **restart**). These saved concentrations replace the workbook initial values.

POC/PIC export and weathering remain fixed; gas exchange and carbonate dissolution/burial respond to the evolving state. The atm reservoir is finite. **Ocean acidification (OA)** here is driven by carbon input; idealized **ocean alkalinity enhancement (OAE)** supplies TA with no direct carbon input.

OA uses the archived IS92a-shaped forcing (scale 0.877, no terrestrial uptake). OAE uses the same time profile, with its own prescribed amount. These are a historical benchmark scenario and a teaching intervention, not present-day forecasts. Runs cover model years 0–3800 with a one-month maximum step; year 1800 is the reporting reference, not a switch from exactly zero input.

After changing a forcing choice, rerun the case-building cell and all subsequent cells. Changing baseline geometry, chemistry or process rates requires a new compatible stationary restart.

Question — specify and predict before running

On your completed schematic from 03, add and label the external input arrow for each experiment. Use these arrows to complete the entry-species/receiving-state fields below and the code choices in Exercise 04.2. State what must match the control. Predict the sign of atm CO₂ and surface pH anomalies in OA and OAE; identify a diagnostic whose opposite sign would challenge your prediction. Revisit this prediction in C1.

Case	External input after year 1800	Entry species / receiving state	Initial state and baseline
Control	None	No forcing connection	Archived restart; fixed benchmark parameters
OA	About 4025 Gt C	From your added arrow on the 03 schematic	Same as control
OAE	10 Pmol TA equivalents	From your added arrow on the 03 schematic	Same as control

The whole-run forcing includes a small pre-1800 tail. The supplied audit reports both intervals. The amounts and units differ, so raw OA/OAE curves do not compare equal-strength interventions.

Instructor answer

OA adds carbon to atmospheric CO₂; OAE adds TA to low-latitude surface TA. Match geometry, chemistry, baseline rates, clock and restart with the control. Expected early OA anomalies are positive atm CO₂ and negative surface pH; OAE is expected to give negative atm CO₂ and positive surface pH. Opposite sustained anomalies in those diagnostics would require checking the choices and revisiting the explanation.

▼ Optional reference — inspect the shared workbook again

The following function displays input records only if you call it. The core uses the specification you already inspected in 03. Numerical baseline inputs stay in `model_definition.xlsx`.

```
# Supplied implementation: optional lookup, not another required table-reading task.
def show_input_reference():
    display(pd.DataFrame(input_tables['OceanReservoirs']).set_index('Box ID'))
    display(pd.DataFrame(input_tables['Atmosphere']).set_index('Box ID'))
    display(pd.DataFrame(input_tables['TransportConnections']).sort_values('Order'))
    display(pd.DataFrame(input_tables['GasExchangeConnections']).sort_values('Order'))
    display(pd.DataFrame(input_tables['ProcessParameters']).set_index('Parameter'))
# To inspect the tables again, run show_input_reference().
```

```
# PDF export: execute the optional workbook lookup in full.
show_input_reference()
```

	Description	Area (m2)	Volume (m3)	Temperature (degC)	Salinity	Pressure (bar)	Initial DIC (umol/kg)	Initial TA (umol/kg)
Box ID								
H_b	High-latitude surface	50000000000000	1760000000000000	2.0	35	17.6	2153	2345
L_b	Low-latitude surface	2850000000000000	2850000000000000	21.5	35	5.0	1952	2288
D_b	Deep ocean	3360000000000000	12900000000000000	2.0	35	240.0	2291	2399

	Total air (mol)	Initial CO2 (ppm)	Role
Box ID			
CO2_At	17786000000000000000	280	Prognostic CO2

	Order	source	sink	flux_id	Parameter	Species
0	1	H_b	D_b	mix_down	mixing	DIC, TA
1	2	D_b	H_b	mix_up	mixing	DIC, TA
2	3	L_b	H_b	thc	thc	DIC, TA
3	4	H_b	D_b	thc	thc	DIC, TA
4	5	D_b	L_b	thc	thc	DIC, TA

	Order	Atmosphere	Surface	Species	Parameter
0	1	CO2_At	H_b	CO2	piston_velocity
1	2	CO2_At	L_b	CO2	piston_velocity

	Value	Unit	Role	Description
Parameter				
thc	25.00000	Sverdrup	Prescribed transport	Water transport on every leg of the overturning loop
mixing	30.00000	Sverdrup	Prescribed transport	Equal exchange in both directions between high latitude and depth
poc_export	200.00000	Tmol/yr	Prescribed export	POC carbon export from low-latitude surface to depth
rain_ratio	0.30000	1	Prescribed ratio	PIC/POC carbon-export ratio at unit pump strengths
weathering_dic	12.00000	Tmol/yr	Prescribed boundary flux	Carbonate weathering: one DIC and two TA equivalents per mole
alpha	0.60000	1	Benchmark coefficient	Carbonate compensation coefficient
z0	-200.00000	m	Benchmark elevation	Carbonate compensation reference elevation
piston_velocity	4.80000	m/d	Prescribed exchange rate	Common gas transfer velocity at both surface boxes
opt_k_carbonic	13.00000	1	Chemistry choice	PyCO2SYS carbonic-acid constant option
opt_ph_scale	3.00000	1	Chemistry choice	PyCO2SYS pH scale option
soft_reference	269.90972	ppm	Reference-state normalization	Archived restart atmospheric CO2 for the optional feedback
soft_half_saturation	280.00000	ppm	Feedback hypothesis	Half-saturation concentration for the soft-tissue response
soft_exponent	1.00000	1	Feedback hypothesis	Exponent for the soft-tissue response
carbonate_reference	341.22511	umol/kg	Reference-state normalization	Archived low-latitude TA minus DIC for the optional feedback
carbonate_half_saturation	300.00000	umol/kg	Feedback hypothesis	Half-saturation concentration for the carbonate response
carbonate_exponent	2.00000	1	Feedback hypothesis	Exponent for the carbonate response

A2. Exercise 04.1: specify the two forcing inventories

Use the amounts in A1's table for the interval after model year 1800.

Question — specify forcing inventories

Assign `oa_target_mol` and `oae_target_mol` in mol C and mol TA equivalents. One Gt is 10^{15} g, one Pmol is 10^{15} mol, and carbon has molar mass 12 g/mol. The supplied scaling preserves the archived shape and its small pre-1800 tail; B1 reports both the post-1800 and whole-run inputs.

```
# Choose and explain: complete marked choices; surrounding machinery is supplied.
# Exercise 04.1: convert the prescribed amounts to the model's mol units.
# BEGIN SOLUTION
oa_target_mol = 4025.0 * 1e15 / 12.0
oae_target_mol = 10.0 * 1e15
# END SOLUTION
# Supplied conversion from inventory to the archived signal's scale factor.
# Keep its exact reference normalization; 4025 Gt is a rounded comparison target.
```

```
np.testing.assert_allclose(oa_target_mol / 1e15, REFERENCE_CARBON_PMOL, rtol=1e-3)
oae_scale = REFERENCE_SCALE * oae_target_mol / (REFERENCE_CARBON_PMOL * 1e15)
```

A3. Exercise 04.2: map the external arrows

Question — implement the experiment specification

Inside each branch, assign `forcing_species` and `forcing_target` from your diagram. Choose from the model's CO₂ and TA species and its atmospheric CO₂ or low-latitude surface TA state. No pH is imposed; it is calculated from the evolving state.

Source represents material outside the boundary. `signal` supplies a time-dependent flux; the native connection attaches it to the chosen state. The control has no such connection. Constructor syntax, waveform scaling and the numerical audit are supplied.

```
# Choose and explain: complete marked choices; surrounding machinery is supplied.
def build_complete_case(forcing=None):
    # Supplied model and stationary restart; pumps remain fixed.
    params = make_pump_variant(base=P) # independent copy with fixed baseline export
    model = initialize_model(params, stop='3800 yr', max_timestep='1 month')
    model.read_state(directory=str(STATE))
    if forcing is None:
        return model
    # Exercise 04.2: set the species and receiving state in both cases.
    if forcing == 'OA':
        # BEGIN SOLUTION
        forcing_species, forcing_target = model.CO2, model.CO2_At
        # END SOLUTION
    elif forcing == 'OAE':
        # BEGIN SOLUTION
        forcing_species, forcing_target = model.TA, model.L_b.TA
        # END SOLUTION
    else:
        raise ValueError(forcing)
    scale = {'OA': REFERENCE_SCALE, 'OAE': oae_scale}[forcing]
    signal = Signal(name='external_input', species=forcing_species, register=model,
                    filename=str(PULSE_FILE), scale=scale)
    source = Source(name='external_source', species=forcing_species)
    connection = Species2Species(source=source, sink=forcing_target,
                                rate='0 mol/yr', signal=signal, id='external_input')
    model.teaching_signal = signal
    model.teaching_connection = connection
    # Preserve the exact solver input for the supplied continuous budget audit.
    model.teaching_signal_time = model.time.copy()
    model.teaching_signal_flux = signal.m.copy()
    if forcing == 'OA':
        model.carbon_signal = signal
    else:
        model.alkalinity_signal = signal
    return model

fixed_cases = {label: build_complete_case(None if label == 'control' else label)
               for label in ('control', 'OA', 'OAE')}
assert fixed_cases['OA'].teaching_connection.sink is fixed_cases['OA'].CO2_At
assert fixed_cases['OAE'].teaching_connection.sink is fixed_cases['OAE'].L_b.TA
```

ESBMTK 0.14.3.1.post0

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B1. Inspect the forcing, then check the runs

The first plot shows the **actual input rates supplied to the solver**, before any response. Separate axes retain mol C versus TA-equivalent units; the third panel divides each rate by its own peak to compare timing only. The vertical line marks year 1800. Follow the rise, peak and decline and inspect whether input has ended when interpreting a later response. Area under a rate curve is an inventory; peak height is not the total input.

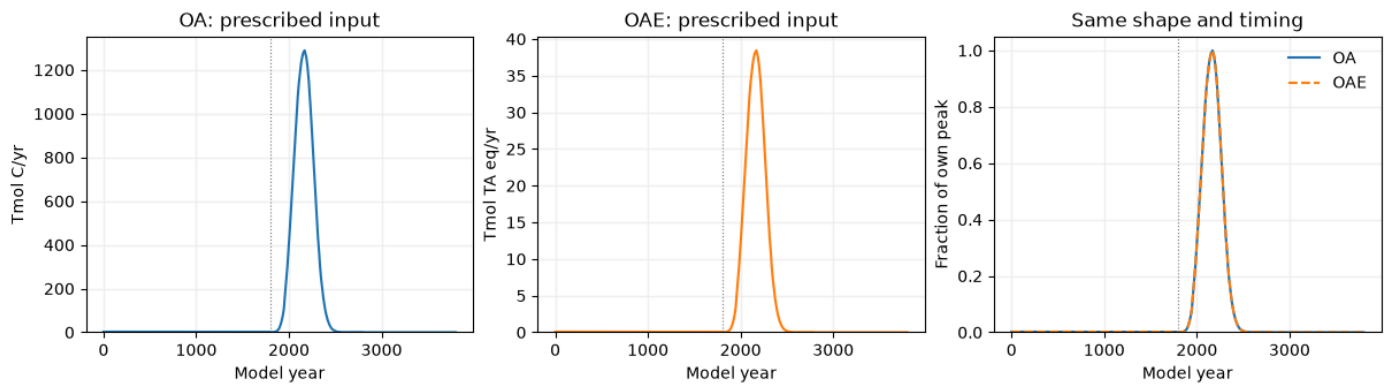
The supplied inventory check then precedes the three model integrations. Reuse 03's active boundary (atm plus dissolved ocn; sediment outside) and its weathering carbon flux W_0 :

$$\frac{dC_{atm+ocn}}{dt} = I_C(t) + W_0 - B_{net}(t), \quad \frac{dA_{ocn}}{dt} = I_A(t) + 2W_0 - 2B_{net}(t).$$

Here I_C is experimental carbon input [mol C/yr] and I_A is experimental TA input [equivalents/yr]. B_{net} is the signed net burial already defined in 03; negative values return sediment material to the active inventories. OA has $I_A = 0$; pure-TA OAE has $I_C = 0$.

Inspect the supplied budget errors once before interpreting results. They compare inventory changes with integrated boundary fluxes, allowing for numerical quadrature; passing does not require constant inventories or establish scientific realism.

```
# Understand and run: inspect prescribed causes before interpreting responses.
from teaching_plots import plot_external_forcings
plot_external_forcings(fixed_cases, pulse_start=PULSE_START)
plt.show()
```



```
# Understand and run: trace this supplied step and interpret its evidence.
from teaching_audits import integrate_forcing_history

forcing_rows = {}
for label, target in (('OA', oa_target_mol), ('OAE', oae_target_mol)):
    case = fixed_cases[label]
    time, flux = case.teaching_signal_time, case.teaching_signal_flux
    total = integrate_forcing_history(time, flux, time[-1])
    after_start = total - integrate_forcing_history(time, flux, PULSE_START)
    np.testing.assert_allclose(after_start, target, rtol=1e-3)
    forcing_rows[label] = {'post-1800 input (Pmol C or TA eq)': after_start / 1e15,
                          'whole-run input (Pmol C or TA eq)': total / 1e15}
display(pd.DataFrame(forcing_rows).T)

for label, case in fixed_cases.items():
    print('Running', label)
    run_model(case)
    postprocess_carbonate_horizons(case)
display(pd.DataFrame({label: audit_complete_model(case, label)
                      for label, case in fixed_cases.items()})).T
```

	post-1800 input (Pmol C or TA eq)	whole-run input (Pmol C or TA eq)
OA	335.357579	340.172309
OAE	10.000047	10.143617

Running control

status=0

message=The solver successfully reached the end of the integration interval.

Execution took 12.55 CPU seconds, wall time = 12.59 seconds

Running OA

status=0

message=The solver successfully reached the end of the integration interval.

Execution took 20.39 CPU seconds, wall time = 20.48 seconds

Running OAE

status=0

message=The solver successfully reached the end of the integration interval.

Execution took 15.08 CPU seconds, wall time = 15.14 seconds

	max carbon error / initial stock	max TA error / initial stock	external input (Pmol C or TA eq)
control	1.084733e-08	2.096911e-08	0.000000
OA	8.335061e-06	1.108447e-06	340.172309
OAE	7.444546e-08	2.116669e-07	10.143617

B2. Interpret matched responses

For each quantity use the **matched anomaly**

$$\Delta Y(t) = Y_{forced}(t) - Y_{control}(t).$$

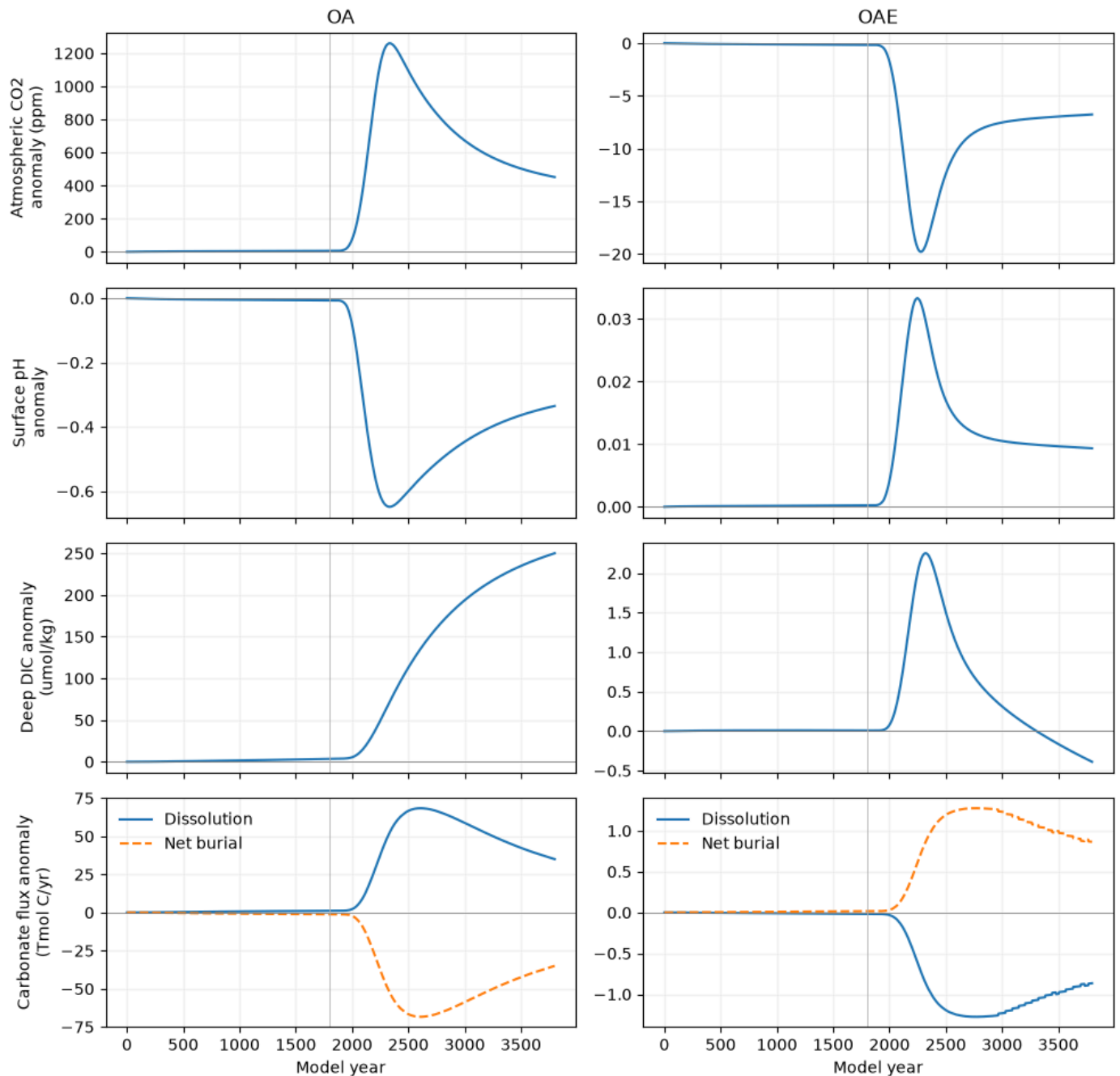
Subtracting the control removes shared baseline drift. It isolates the response to the prescribed input within this model; it does not remove numerical error or validate omitted processes.

Read the atm CO₂ and low-latitude surface pH anomalies first, then deep DIC and dissolution/net burial. Compare their signs and turning points with the forcing in B1: a response maximum need not coincide with the input-rate maximum. Surface chemistry, gas exchange, transport and sediment adjustment act together, with different response times.

Compare mechanisms and timing, not raw amplitudes: the two columns have different vertical scales, input species and total amounts. The overlaid normalized forcings establish common timing only; they do not make the interventions equally strong. Use B2 for C1–C2 and the depth/flux panels below for C3–C4.

```
# Understand and run: trace this supplied step and interpret its evidence.
from teaching_plots import plot_matched_responses
plot_matched_responses(fixed_cases, pulse_start=PULSE_START)
plt.show()
```

Carbon versus alkalinity input: forced minus matched control



B3. Read critical depths and carbonate preservation

These complete figures reuse the same runs. Solid lines are model results; OA's dotted lines are archived digitizations of the published benchmark. Use them to assess reproduction, while B2's matched anomalies identify the forcing response.

Panels	Read for
a / b / c	DIC, TA and pH: surface-to-deep chemical change
d / f	Air-sea transfer (positive into the ocn) and atm CO ₂
e / h	Critical depths, dissolution and net burial: carbonate preservation
g	The forcing from B1, repeated here as a timing reference

The three **critical depths** answer different questions. Calcite saturation Ω compares the calcium–carbonate ion product with its equilibrium solubility product; $\Omega < 1$ means undersaturation.

Depth	Physical meaning	What controls it here?
z_{sat} : saturation horizon	Water has $\Omega_{calcite} = 1$; deeper water is undersaturated	Current deep carbonate chemistry and pressure-dependent solubility
z_{cc} : carbonate compensation depth (CCD)	Dissolution balances the arriving modern $CaCO_3$ rain; none survives for burial	Current chemistry, dissolution kinetics and prescribed rain
z_{snow} : snowline	Deepest boundary of existing reactive carbonate sediment	Past deposition and dissolution; erosion takes time

Thus undersaturated water does not imply immediate disappearance of all sediment carbonate. This distinction follows [Boudreau et al. \(2010\)](#), *Carbonate compensation dynamics*. Depth is positive downward in these definitions, but **panel e plots elevation** ∴ upward means shallower. The lower limit of the implemented saturation horizon is 200 m; reaching that bound is not evidence that every surface water is undersaturated.

Supplied snowline rule: shoaling and deepening differ

The CCD describes whether **today's carbonate rain** survives; the snowline describes where **carbonate sediment is already present**. Use the following model rules to interpret panels e/h. You are not expected to look up or derive the sediment code.

- **CCD shoaling (the main OA response):** the CCD moves into shallower water, leaving previously deposited carbonate below it. Modern rain there dissolves completely, but the existing sediment stock takes time to dissolve. The snowline therefore remains deeper than the CCD and moves upward more slowly: $z_{snow} > z_{cc}$. Their separation records sediment left from earlier conditions.
- **CCD deepening (the main OAE response):** newly arriving carbonate can survive at greater depths. The model lets the snowline follow the deepening CCD effectively instantly on these century-scale plots: $z_{snow} \approx z_{cc}$. This is a supplied assumption about the onset of preservation; it does not simulate the time needed to build a substantial carbonate-rich sediment layer. Numerical tracking can leave a small lag.

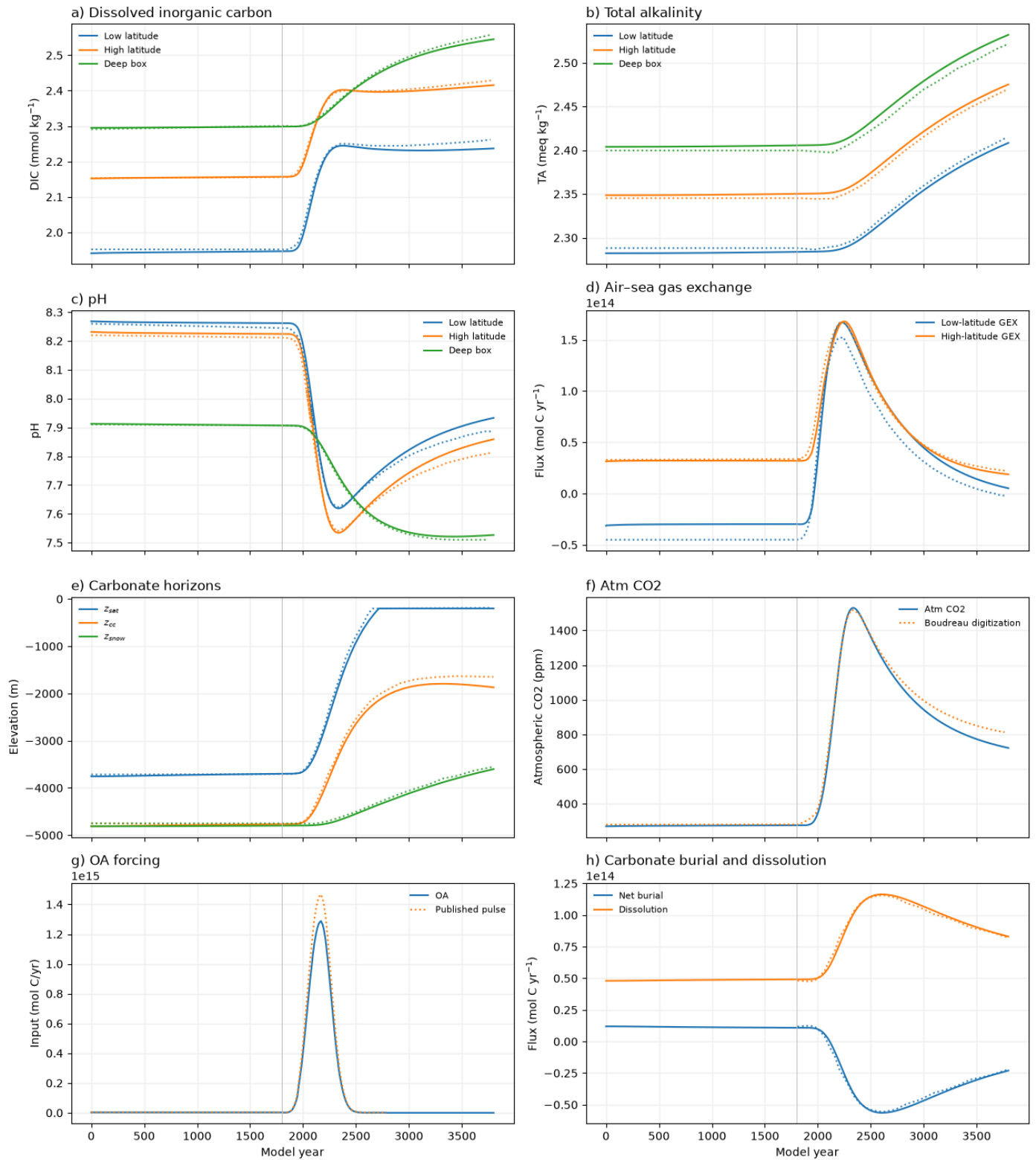
Here **instant** means relative to the moving CCD, not relative to the external TA input: transport and deep-water chemistry still take time to change. The rule follows the direction of depth change, not the OA/OAE label; a later shoaling CCD can leave a lagging snowline even in OAE. Read e together with h to connect depth separation to dissolution and net burial.

Chemical carbonate compensation means changing dissolution/preservation at fixed carbonate rain. It changes dissolved TA, unlike acid–base repartitioning among dissolved species. POC and PIC exports remain prescribed in all three cases: changing deep DIC alone cannot establish changing biological export.

The benchmark's transient depth separation and sediment response are discussed in [Boudreau et al. \(2010\)](#), *Ongoing transients in carbonate compensation, section 4.2*. Read the selected panels for the four answers below; no sediment-equation derivation or panel-by-panel report is required.

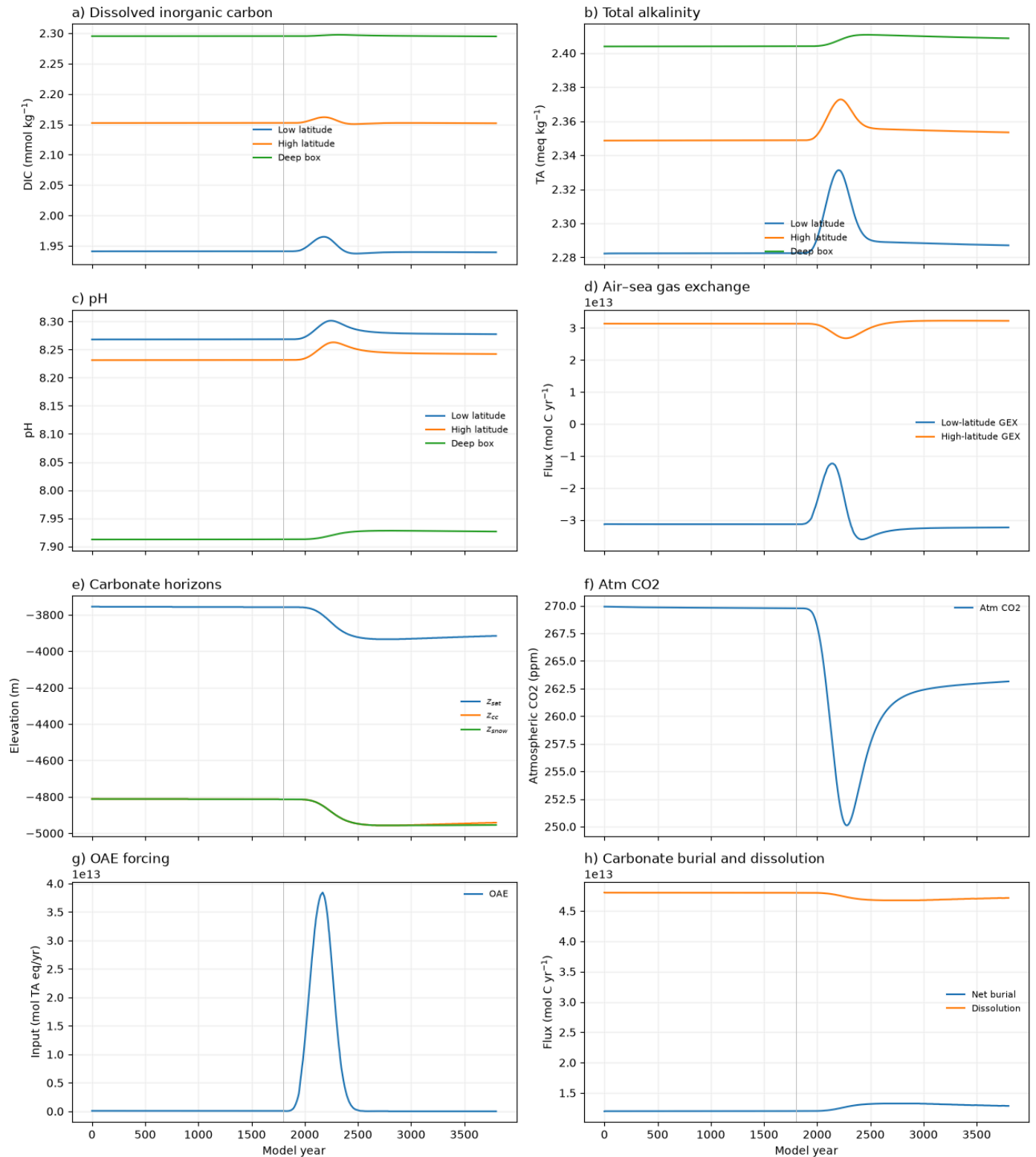
```
# Understand and run: trace this supplied step and interpret its evidence.
from teaching_plots import plot_figure4
plot_figure4(fixed_cases['OA'], 'OA', reference=True,
             digitized=DIGITIZED, pulse_start=PULSE_START)
plt.show()
```

OA: complete Boudreau-like model with constant pumps



```
# Understand and run: trace this supplied step and interpret its evidence.
plot_figure4(fixed_cases['OAE'], 'OAE', pulse_start=PULSE_START)
plt.show()
```

OAE: complete Boudreau-like model with constant pumps



C. Explain the scientific results in four short answers

Use two or three sentences per answer, referring to a curve and an approximate model year or interval. Read approximate values from the supplied plots; no extra calculations or runs are required.

Question — mechanisms, critical depths and OA/OAE asymmetry

1. **Forcing and response:** revisit your sign prediction using B2. Compare the input-rate peak with the atm/surface response and one delayed deep response. Does declining input mean the perturbation is over?
2. **Carbon uptake and TA:** explain OAE's atm CO₂ response without a direct carbon input. Contrast TA-conserving dissolved-species repartitioning with the changing ocn TA in OA; use the net-burial term in B1 and panel h as evidence.
3. **Critical depths and memory:** use B3's supplied rules and panels e/h. During OA shoaling, locate z_{sat} , the CCD and the snowline at about year 2800: what survives between the CCD and snowline, and how does this explain the snowline lag and net-burial sign? During OAE deepening, explain the near-overlap of CCD and snowline. Does this show instantaneous deep-ocean adjustment or instantaneous formation of a thick sediment layer? Distinguish the plotted result from the model assumption.
4. **Asymmetry and limits:** identify one difference beyond opposite response signs. Separate unequal forcing amounts/entry species from a chemical or sediment mechanism. Why can these runs establish neither an OAE amount that cancels OA nor stronger biological export from higher deep DIC? State what the late curves and benchmark overlay do (and do not) establish.

Instructor answer

1. Both input rates peak near model year 2170; OA's atm CO₂ maximum/surface pH minimum follow near 2300–2350, while its deep DIC is still rising at 3800. OAE lowers atm CO₂ and raises surface pH; deep DIC initially rises too, then declines and becomes slightly negative relative to control late in the run. Declining positive input still adds material, and adjustment persists after input ends; a turning point is not equilibrium.
2. Adding TA shifts dissolved carbonate speciation and lowers aqueous CO₂, driving internal transfer from atm to ocn without external carbon addition. Dissolved acid–base repartitioning itself conserves TA; OA's enhanced dissolution/reduced net burial retains or returns TA to the active ocn at two equivalents per mole of carbonate carbon (h: burial is negative by about 2300). In OAE, increased net burial instead removes part of the added TA and carbon, weakening sustained uptake; atm loss need not equal ocn gain because this sediment boundary responds.
3. **OA shoaling:** near year 2800, $z_{sat} \approx 0.2$ km, $z_{cc} \approx 2.0$ km and $z_{snow} \approx 4.3$ km; modern rain cannot survive below the CCD, but previously deposited carbonate remains down to the snowline and must dissolve before that boundary can retreat, producing a lag and a memory of earlier conditions. Dissolution of this old stock lets total dissolution exceed the 60 Tmol C/yr modern rain, so net burial becomes negative. **OAE deepening:** the CCD and snowline move together toward about 5.0 km because the model assumes an effectively immediate extension of the preservation boundary as rain starts surviving deeper, with less dissolution and more burial. Their near-overlap is therefore consistent with the supplied rule, not evidence that deep-water chemistry responds instantly to surface TA or that a thick sediment layer forms instantly; the small later separation in OAE appears when its CCD starts shoaling again.
4. After roughly 2800, OA deep DIC continues rising whereas OAE deep DIC declines; enhanced dissolution supplies carbon/TA in OA, while extra net burial in OAE removes them and reduces retained alkalinity. The amplitudes alone cannot establish intrinsic asymmetry because the inputs differ in amount, species and receiving box; nonlinear chemistry and sediment history give additional reasons not to expect mirror images. Fixed POC/PIC export excludes stronger biological export as the explanation of higher deep DIC. These separate perturbations do not test cancellation, late adjustment is not proven stationarity, and the OA overlay supports model reproduction rather than independent observational validation.

You have completed 04. Keep your forcing choices, checked budgets and four interpretations here.